

Breast Cancer Enhanced Dataset Analytics Report

Automated Kaggle analysis for diagnosis patterns, tumor feature distributions, risk scoring, predictive modeling, clustering and anomaly detection

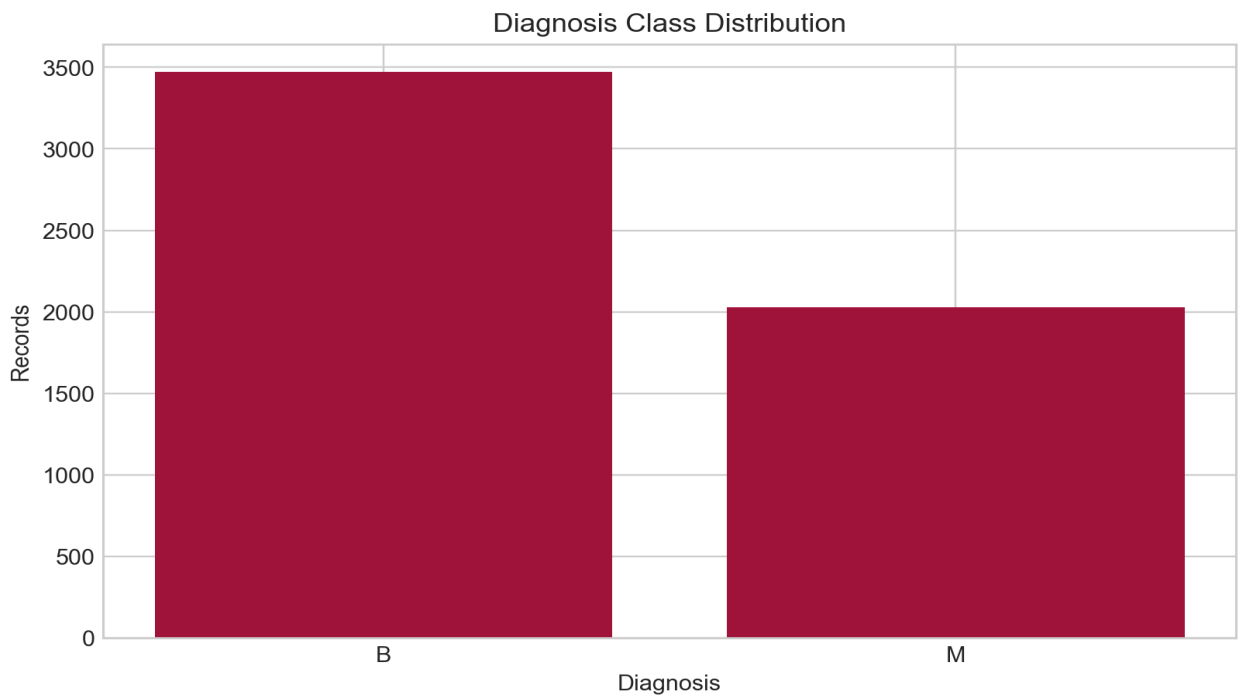
This report analyzes the uploaded dataset **breast_cancer_enhanced_dataset.csv**. It contains **5,500** observations and **18** columns after preprocessing.

Executive Summary Metrics

Metric	Value	Interpretation
Rows	5,500	Total usable observations after diagnosis cleaning
Columns	18	Total available fields including engineered features
Numeric predictors	15	Candidate modeling variables
Benign cases	3,470	Diagnosis coded as benign
Malignant cases	2,030	Diagnosis coded as malignant
Malignancy rate	36.9%	Share of malignant diagnosis records
Average malignancy risk score	29.832	Mean engineered risk-score value

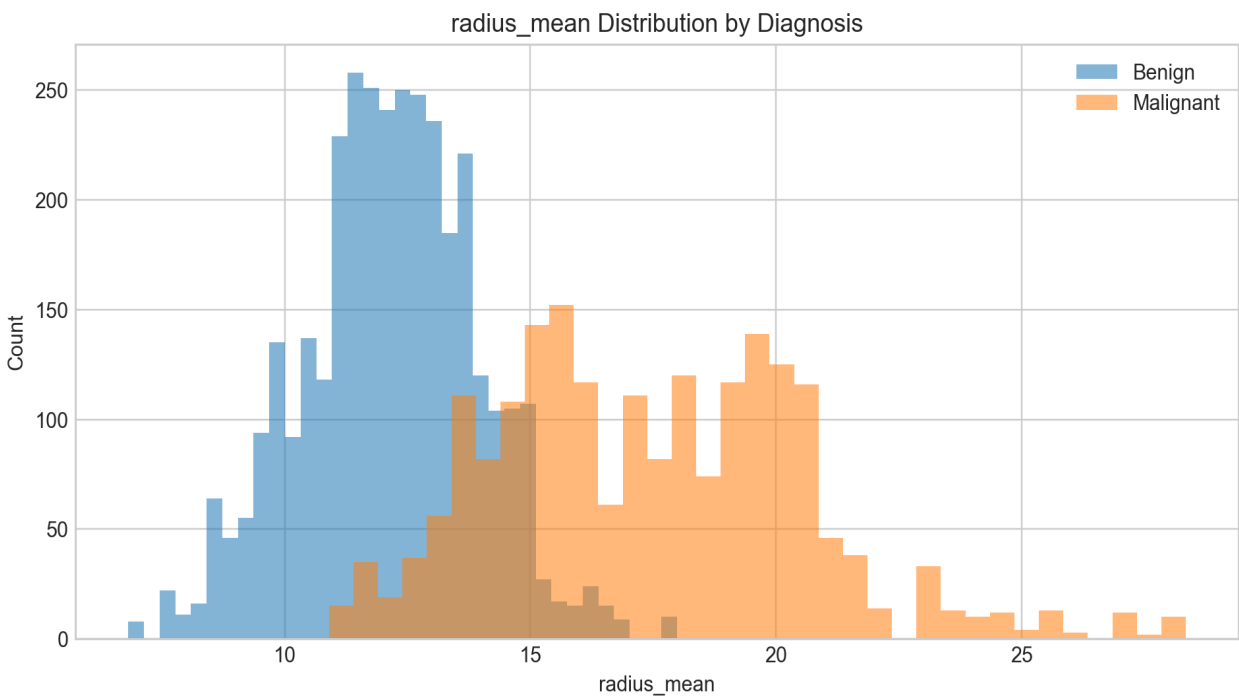
Diagnosis Distribution and Core Tumor Measurements

Diagnosis Class Distribution



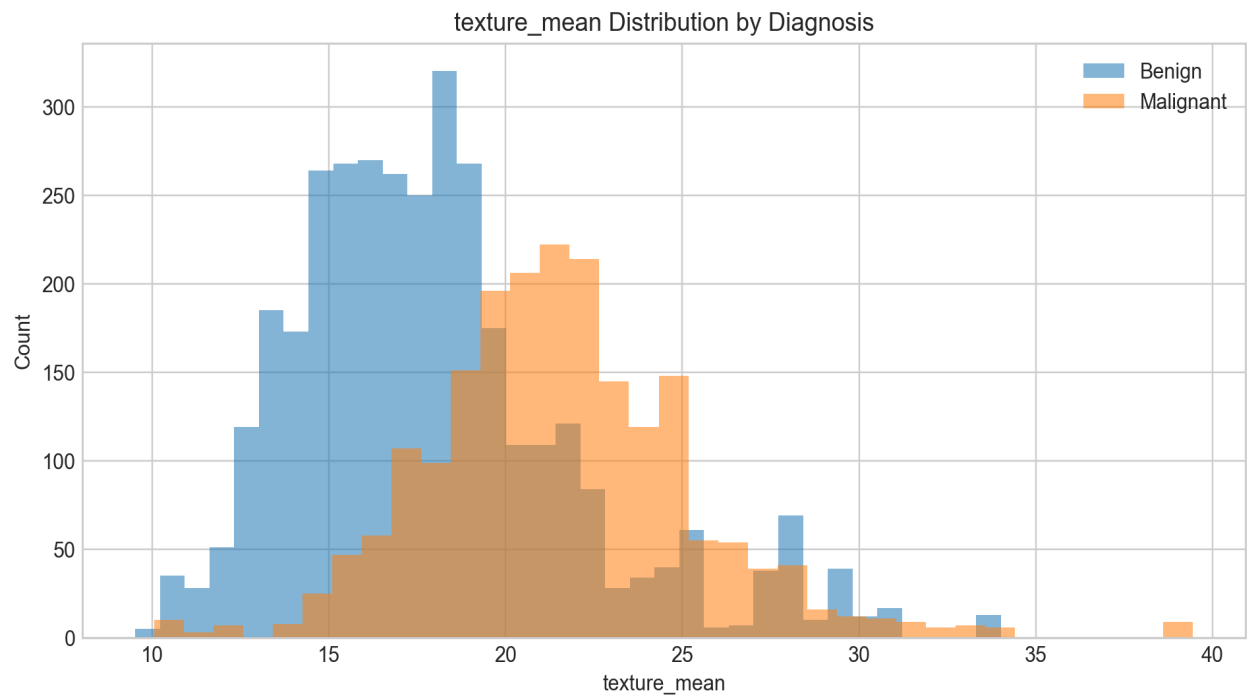
The dataset contains **3,470** benign and **2,030** malignant cases.

radius_mean Distribution by Diagnosis



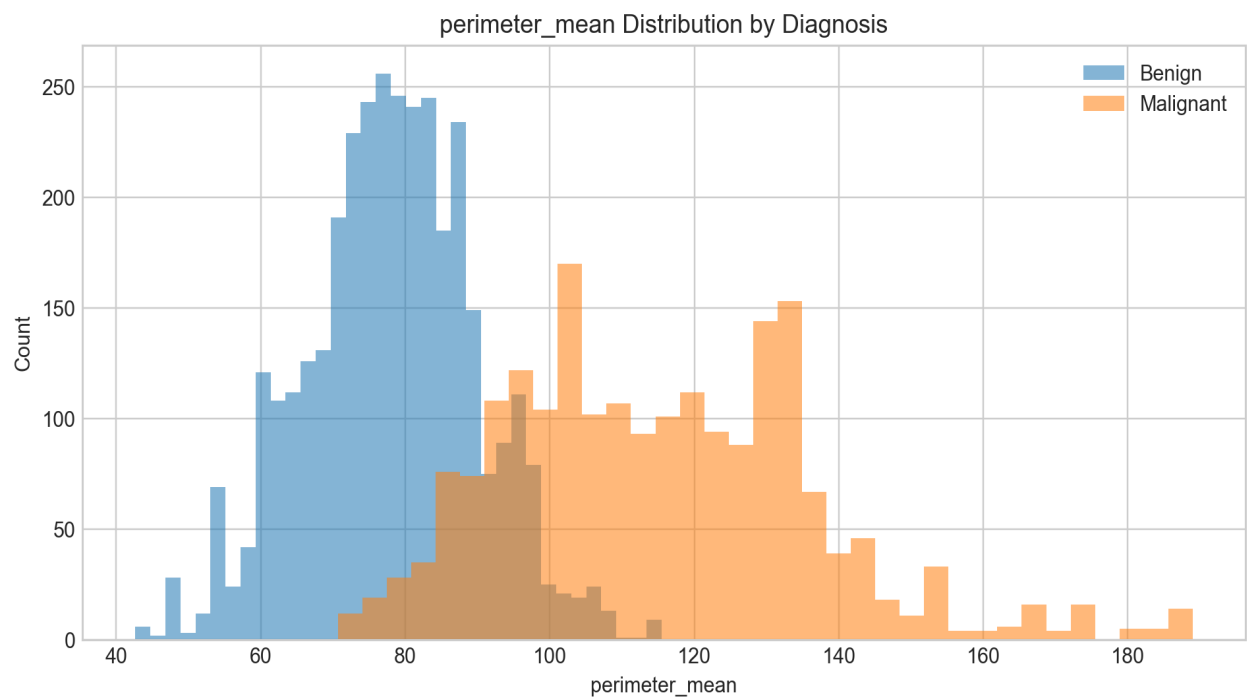
Mean **radius_mean** is **12.148** for benign cases and **17.424** for malignant cases.

texture_mean Distribution by Diagnosis



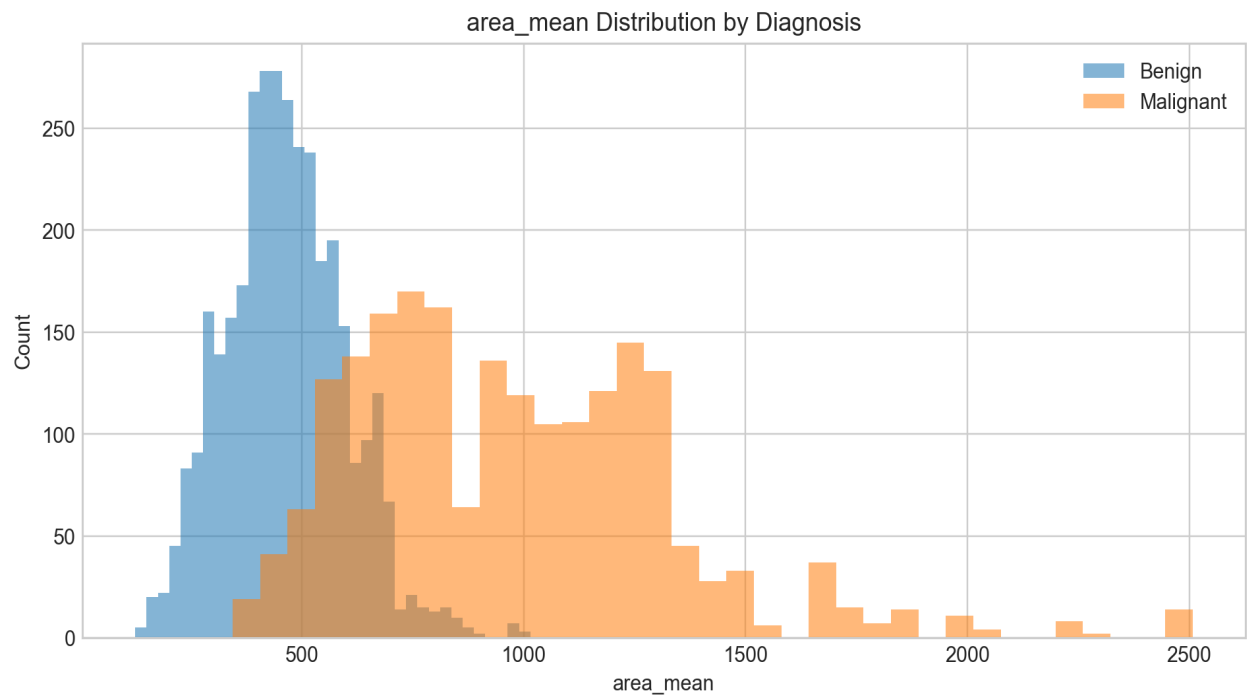
Mean **texture_mean** is **18.001** for benign cases and **21.567** for malignant cases.

perimeter_mean Distribution by Diagnosis



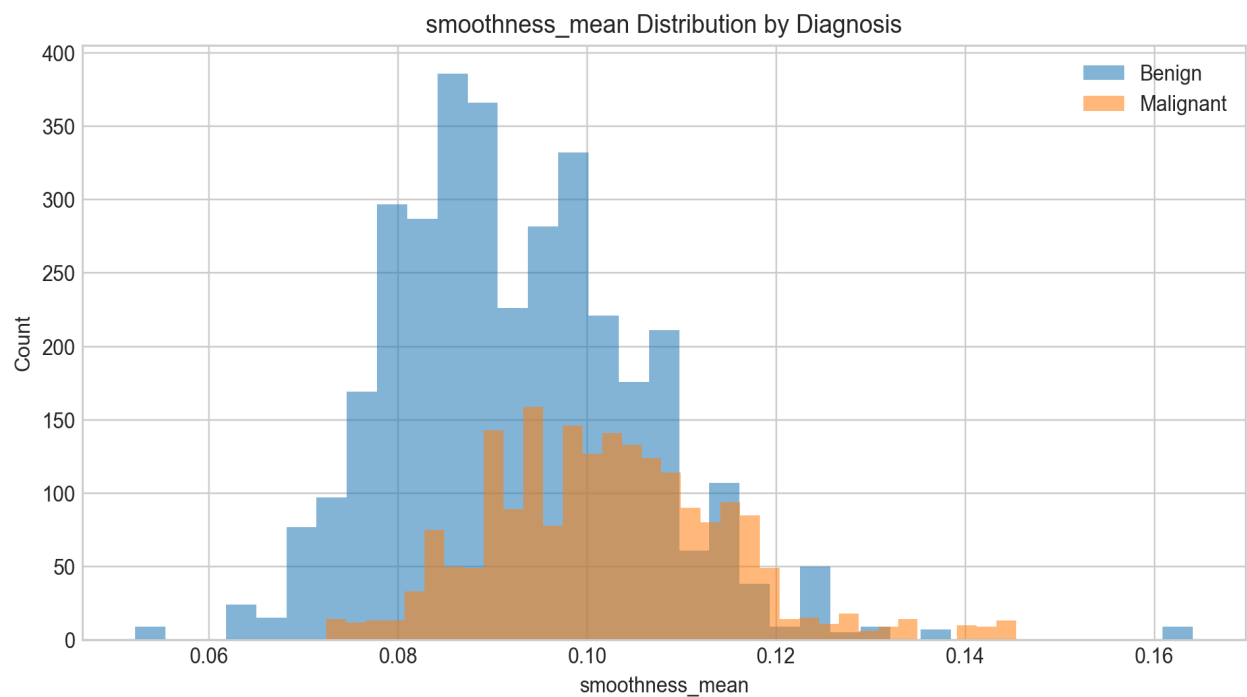
Mean **perimeter_mean** is **78.110** for benign cases and **115.065** for malignant cases.

area_mean Distribution by Diagnosis



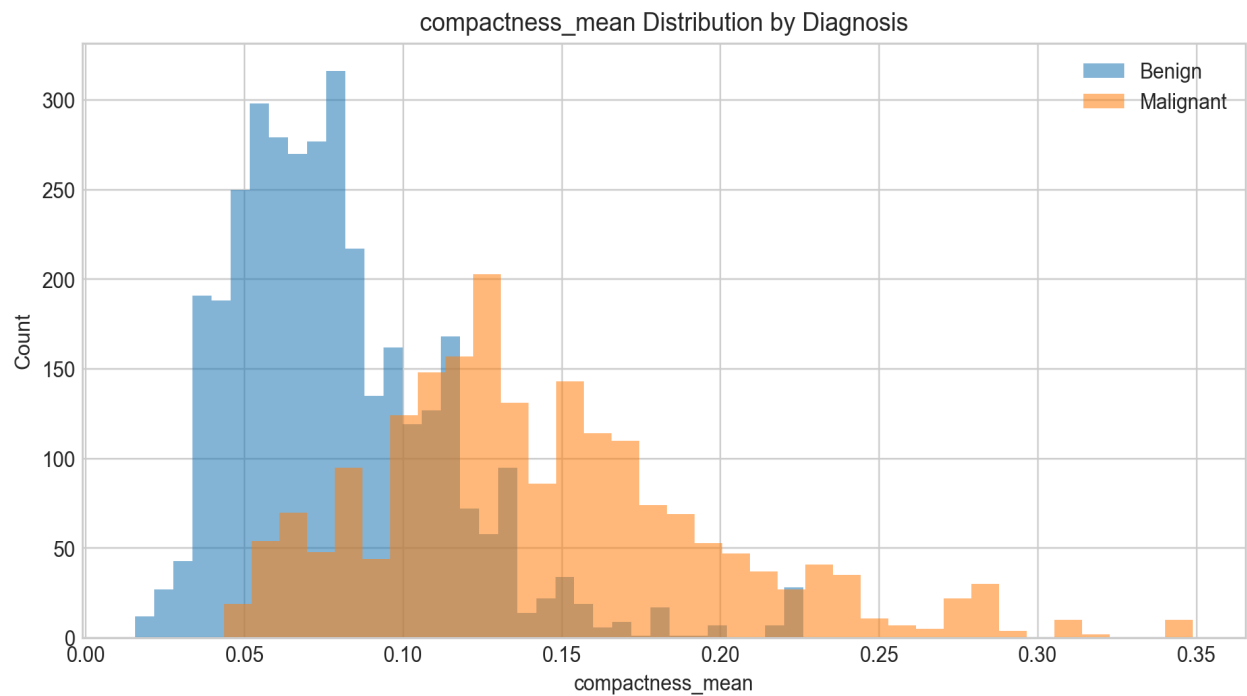
Mean **area_mean** is **463.111** for benign cases and **973.071** for malignant cases.

smoothness_mean Distribution by Diagnosis



Mean **smoothness_mean** is **0.092** for benign cases and **0.102** for malignant cases.

compactness_mean Distribution by Diagnosis



Mean **compactness_mean** is **0.079** for benign cases and **0.144** for malignant cases.

Largest Diagnosis-Level Feature Differences

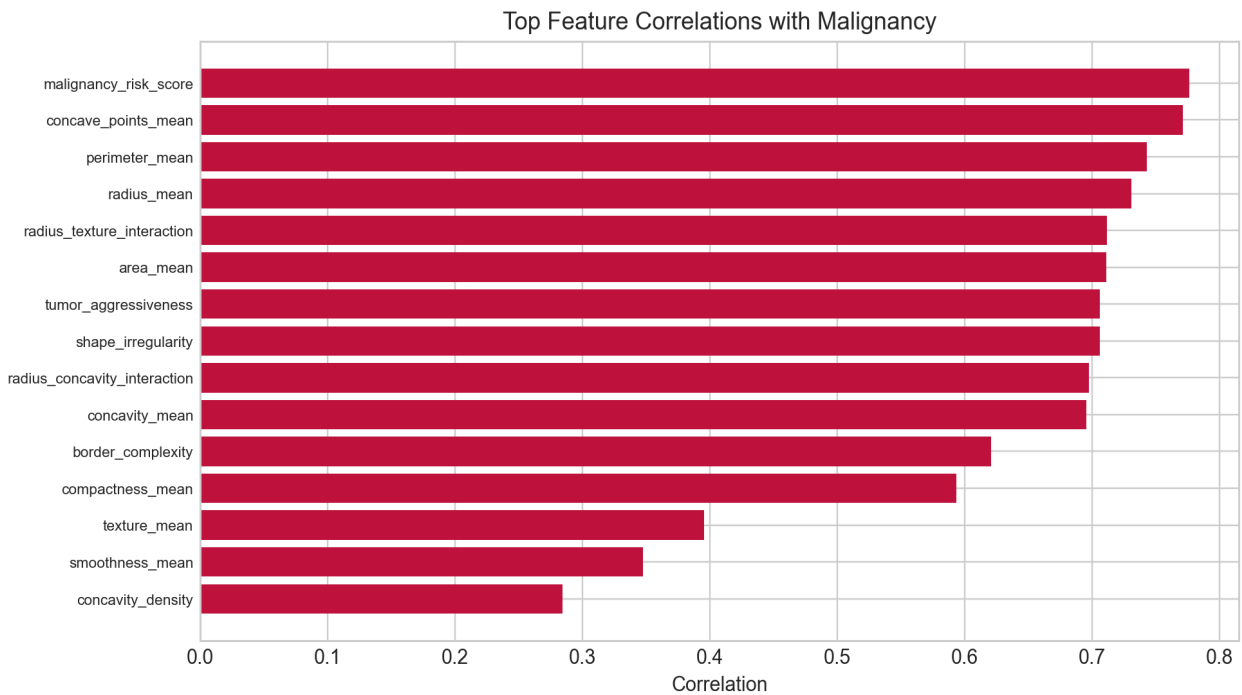
Feature	Benign Mean	Malignant Mean	Difference	Std Difference
malignancy_risk_score	24.320	39.254	14.934	1.609
concave_points_mean	0.025	0.087	0.062	1.598
perimeter_mean	78.110	115.065	36.955	1.539
radius_mean	12.148	17.424	5.277	1.514
radius_texture_interaction	218.246	376.811	158.565	1.475
area_mean	463.111	973.071	509.960	1.474
shape_irregularity	0.149	0.389	0.240	1.464
tumor_aggressiveness	0.050	0.130	0.080	1.464
radius_concavity_interaction	0.552	2.872	2.319	1.445
concavity_mean	0.045	0.158	0.113	1.441
border_complexity	0.002	0.016	0.015	1.286
compactness_mean	0.079	0.144	0.065	1.229
texture_mean	18.001	21.567	3.567	0.820
smoothness_mean	0.092	0.102	0.010	0.720
concavity_density	0.000	0.000	0.000	0.589

Correlation and Feature Relationship Analysis

Top Correlations with Malignant Diagnosis

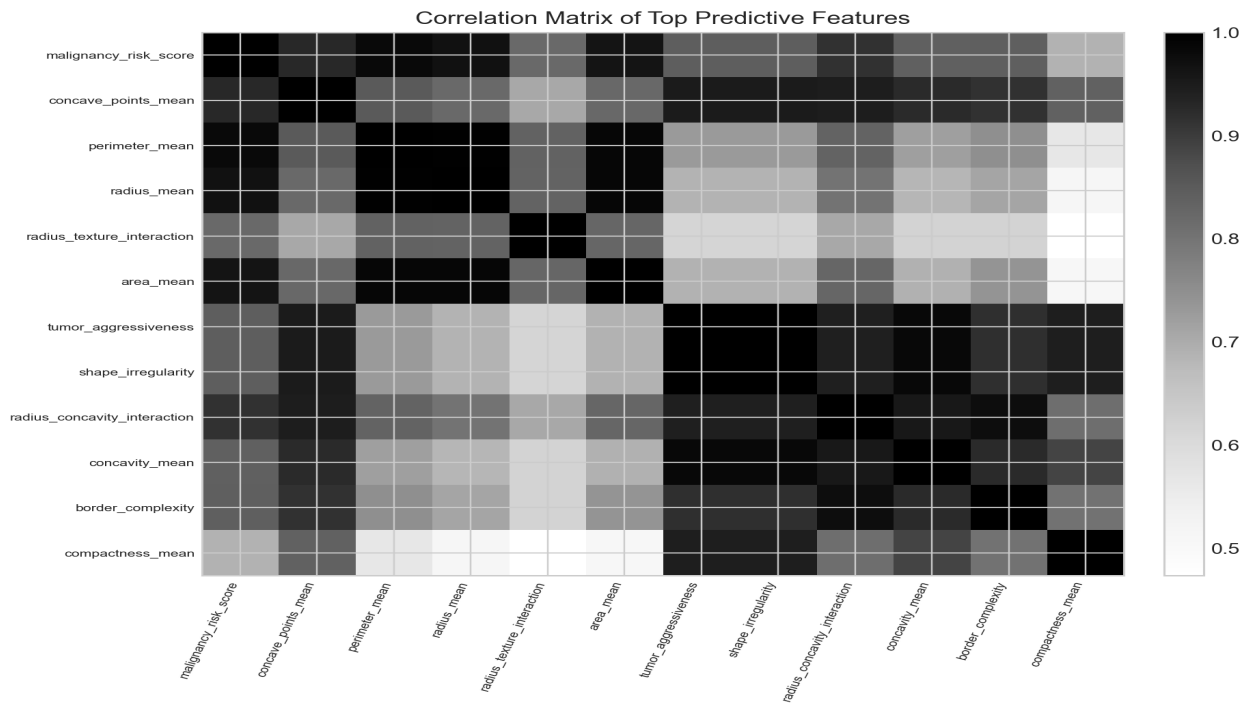
Feature	Correlation with Target
malignancy_risk_score	0.7764
concave_points_mean	0.7713
perimeter_mean	0.7427
radius_mean	0.7308
radius_texture_interaction	0.7117
area_mean	0.7112
tumor_aggressiveness	0.7063
shape_irregularity	0.7063
radius_concavity_interaction	0.6973
concavity_mean	0.6953
border_complexity	0.6208
compactness_mean	0.5932
texture_mean	0.3956
smoothness_mean	0.3472
concavity_density	0.2841

Top Feature Correlations with Malignancy



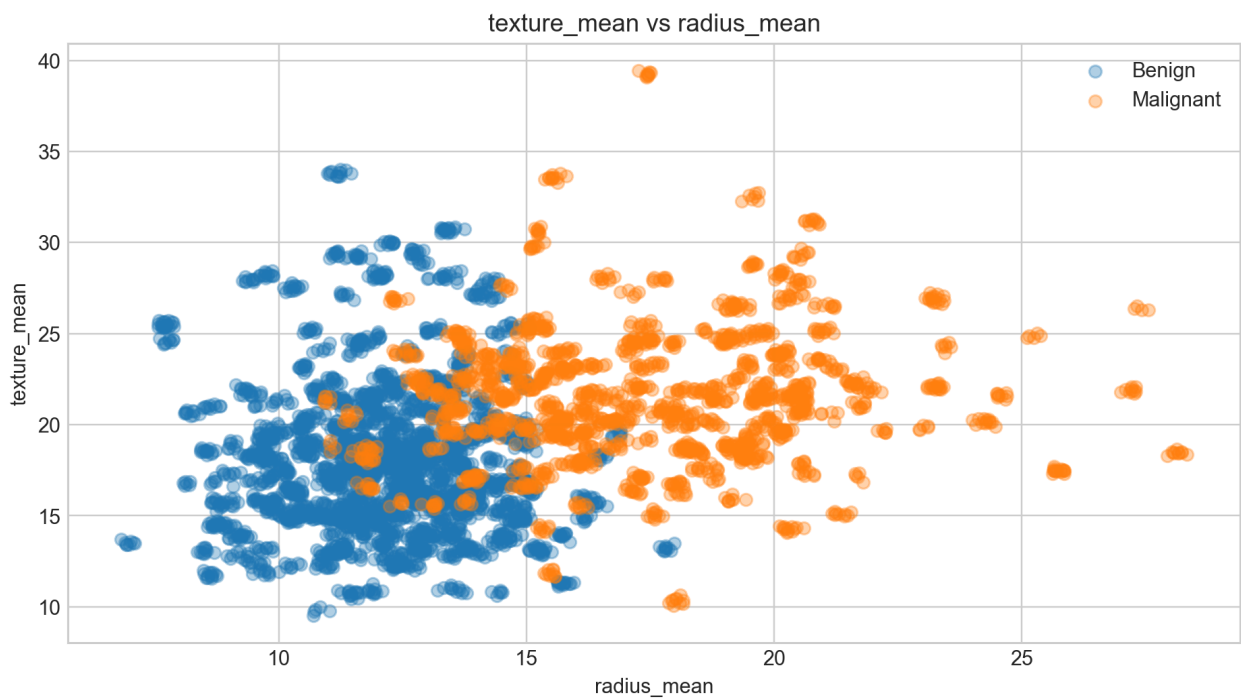
The strongest absolute linear association with malignancy is **malignancy_risk_score** with correlation **0.7764**.

Top Feature Correlation Matrix



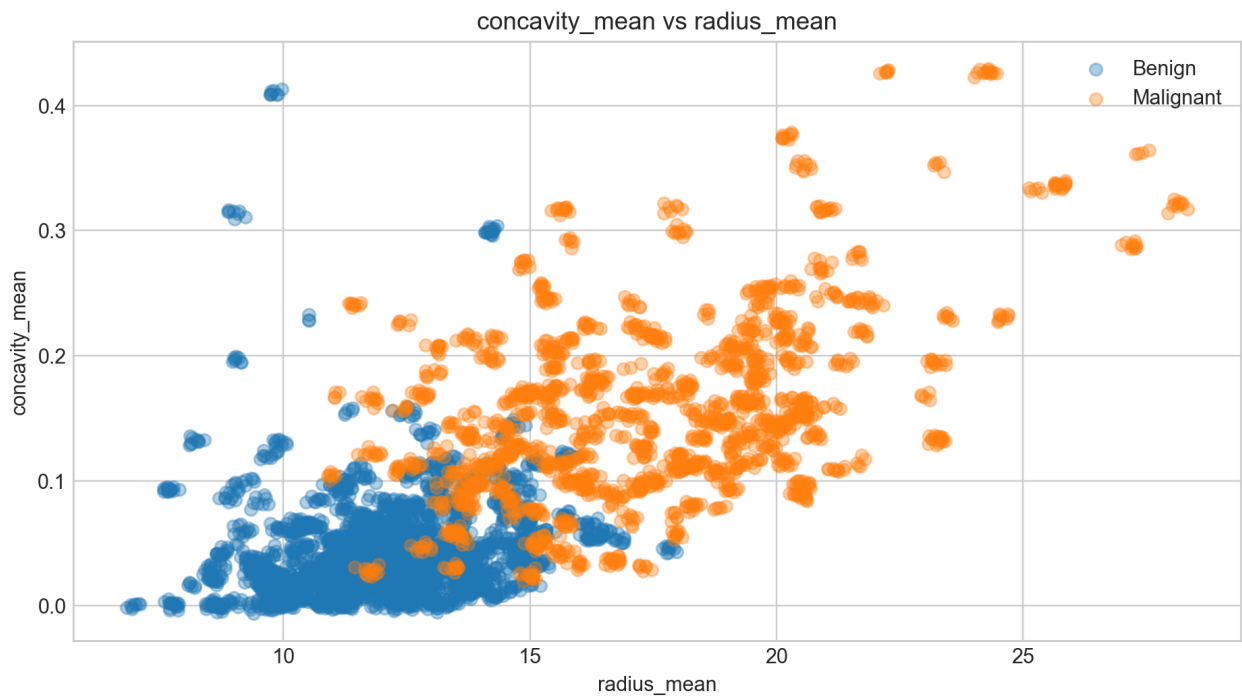
The heatmap highlights redundant or strongly related tumor measurements that may carry overlapping diagnostic information.

texture_mean vs radius_mean



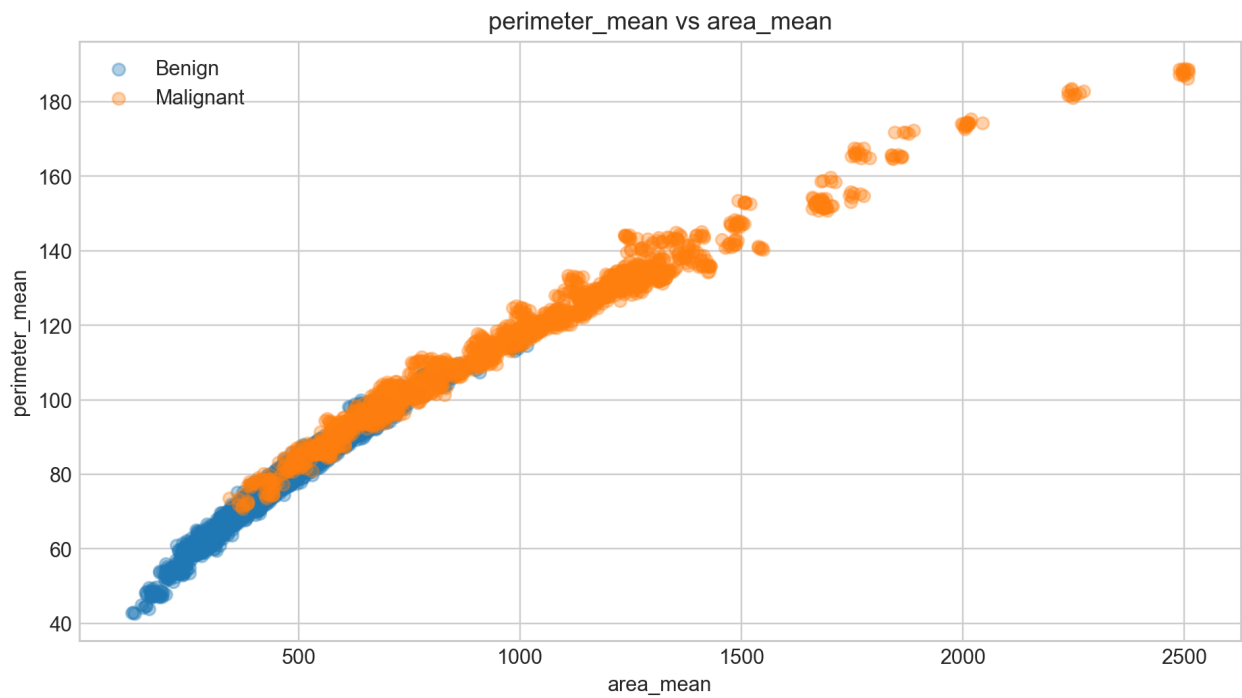
This scatter plot compares **radius_mean** and **texture_mean** across benign and malignant records.

concavity_mean vs radius_mean



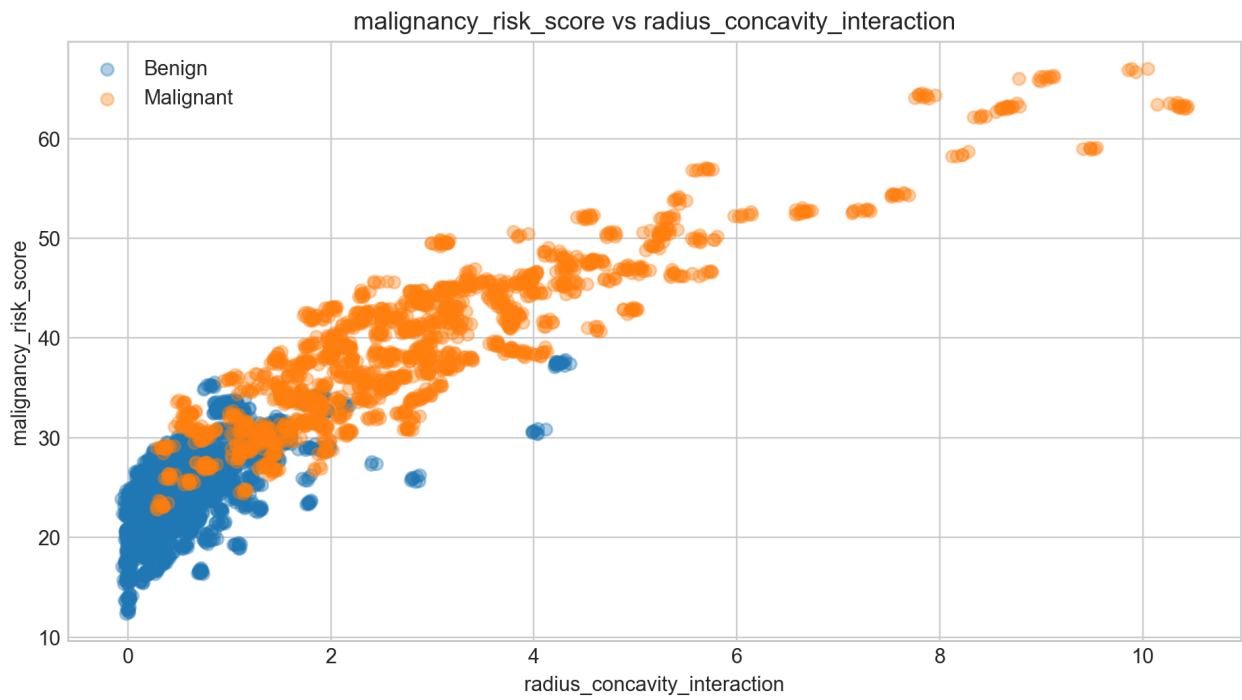
This scatter plot compares **radius_mean** and **concavity_mean** across benign and malignant records.

perimeter_mean vs area_mean



This scatter plot compares **area_mean** and **perimeter_mean** across benign and malignant records.

malignancy_risk_score vs radius_concavity_interaction



This scatter plot compares **radius_concavity_interaction** and **malignancy_risk_score** across benign and malignant records.

Predictive Modeling

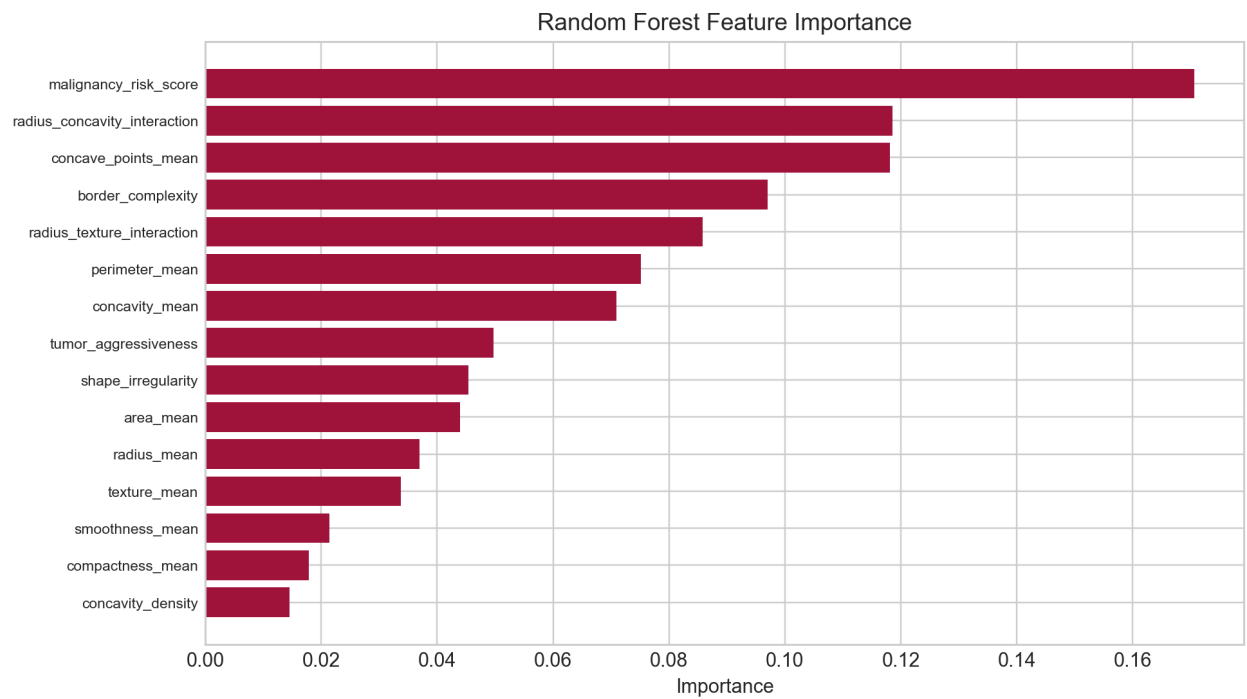
Diagnosis Classification Model Performance

Model	Accuracy	ROC AUC
Random Forest	0.9934	0.9999
Gradient Boosting	0.9868	0.9995

Random Forest Confusion Matrix

	Predicted Benign	Predicted Malignant
Actual Benign	761	2
Actual Malignant	6	441

Diagnosis Classification Feature Importance

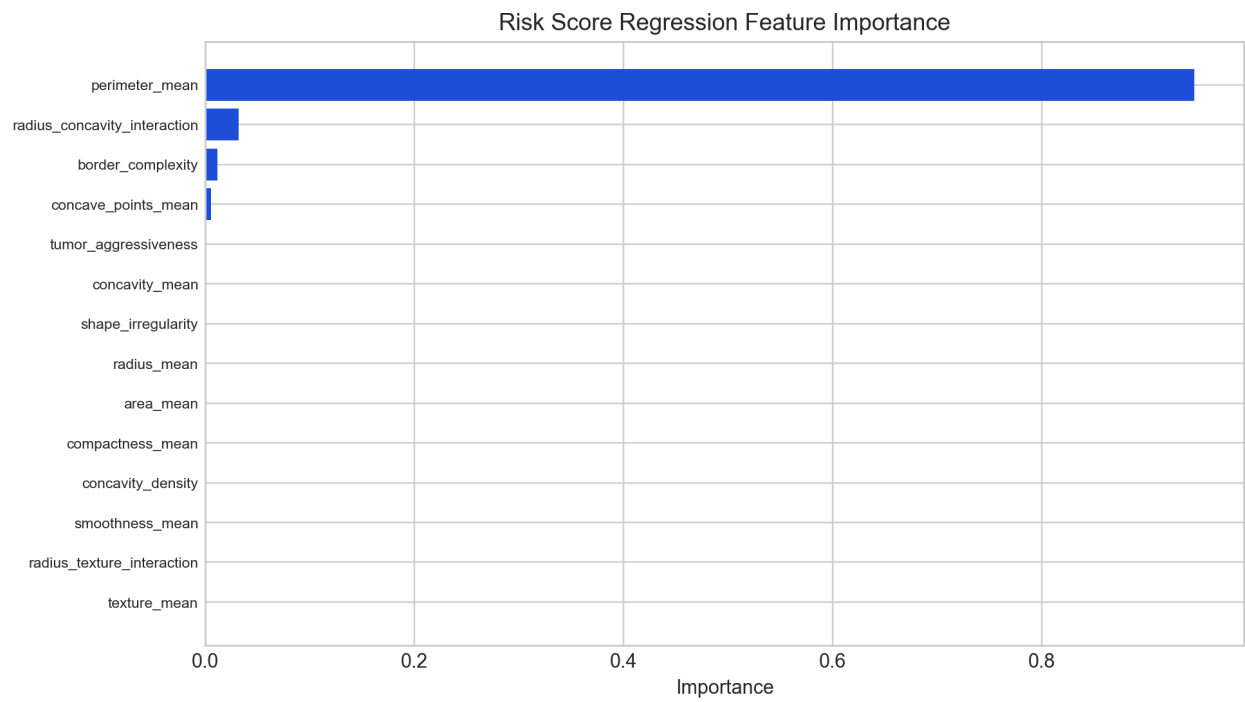


The strongest model feature is **malignancy_risk_score**. The best classifier is **Random Forest** with ROC AUC **0.9999**.

Malignancy Risk Score Regression Performance

Model	RMSE	R2
Random Forest Regressor	0.1417	0.9998
Gradient Boosting Regressor	0.2441	0.9993

Risk Score Feature Importance



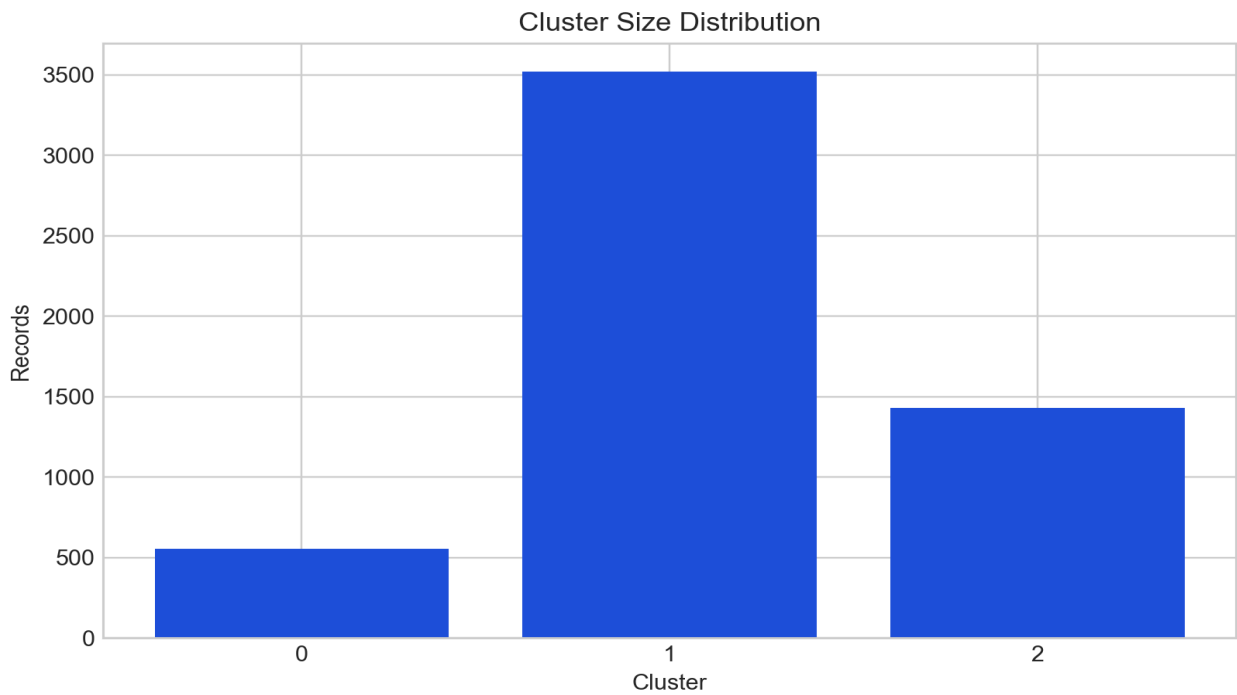
The best risk-score model is **Random Forest Regressor** with RMSE **0.1417** and R2 **0.9998**.

Clustering and Anomaly Detection

Tumor Feature Cluster Summary

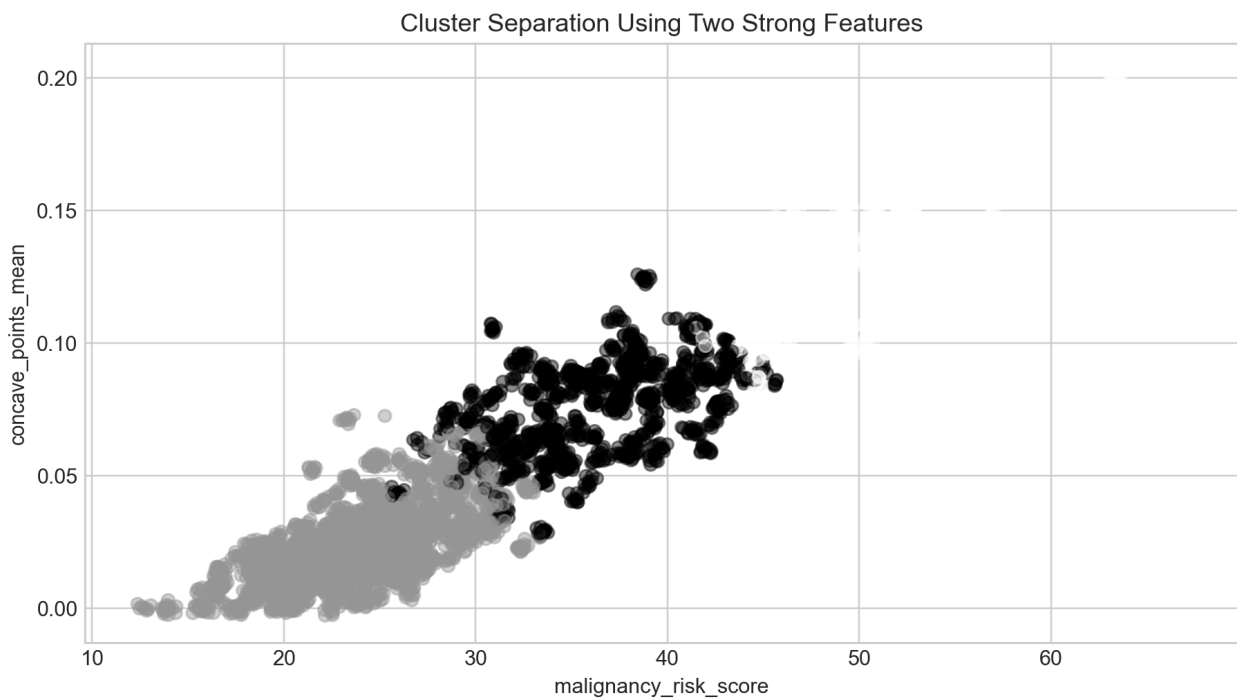
Cluster	Records	Malignancy Rate
0.0	553.0	1.0
1.0	3518.0	0.06907333712336555
2.0	1429.0	0.8635409377186843

Tumor Cluster Size Distribution



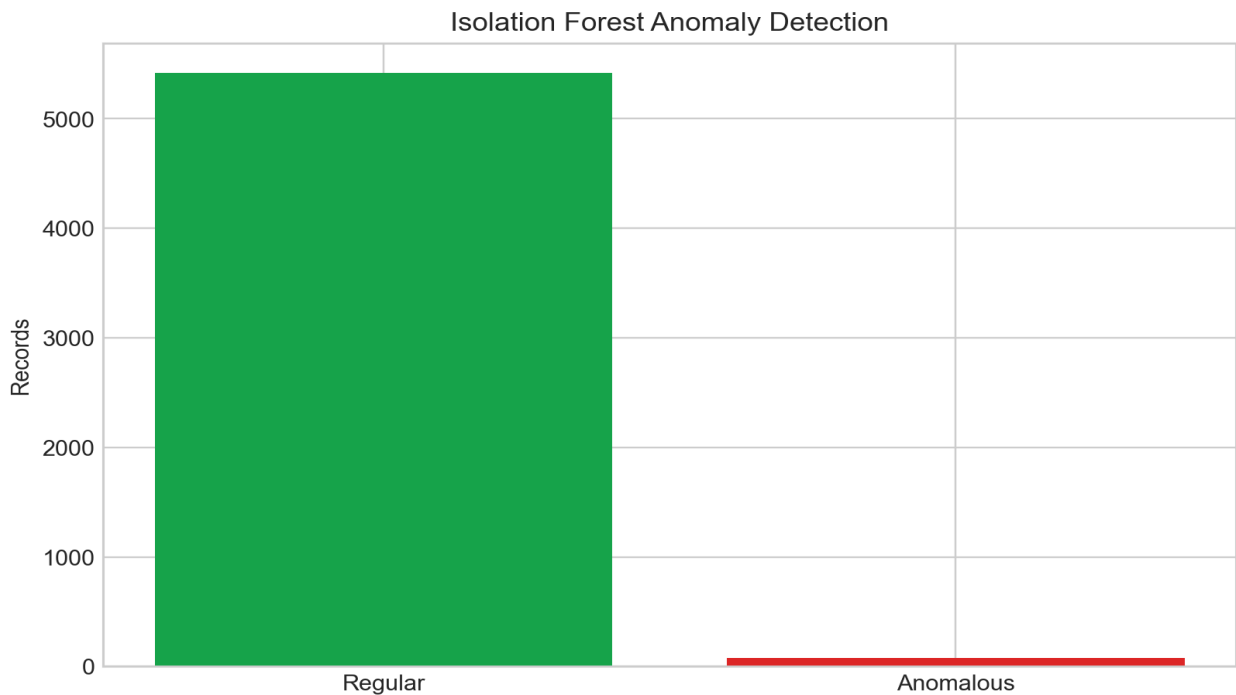
KMeans grouped records into **3** tumor-feature clusters.

Two-Feature Cluster Separation



The chart shows cluster structure using **malignancy_risk_score** and **concave_points_mean**.

Anomaly Detection Summary



Isolation Forest flagged 80 unusual records based on top diagnostic features.

Sample Potentially Anomalous Records

Diagnosis	malignancy_risk_score	concave_points_mean	perimeter_mean	radius_mean	radius_texture_interaction	area_mean
M	63.12543868314673	0.20096120132406	165.51084642422265	24.341384465165152	491.48086089296686	1748.7636548369007
M	62.35439787811026	0.1842099180199607	172.4375164206787	25.247883717306035	625.2533735538498	1888.768864040015
M	64.07435622665564	0.1877681289879994	181.81228423339607	27.18572222021425	592.1864268100516	2236.7714094019634
M	66.15195243446821	0.1584922459119656	188.33103400734447	28.22665868912541	521.5109832451498	2496.6061806587463
M	66.21022027736026	0.1602580264845799	188.73004248555708	28.134575107280355	525.4224504641178	2507.9338647305985
M	64.20770504941505	0.188945298120473	182.0379142656478	27.220856112477364	600.7115003562967	2250.4630067647263
M	65.90480027213674	0.1579634587592727	187.79868643157928	28.037554483248137	518.8379818877486	2507.9823553397123
M	54.292496534742966	0.161867357271718	143.49303891538233	20.12577111397413	479.5598389369204	1247.115872378145
M	62.2269114181856	0.1845197909820876	171.75214990389912	25.307891227979564	633.7361997911602	1866.14641219859
M	64.16835934120418	0.1886950305778455	181.77642902693628	27.24123120926532	595.3482814479855	2241.34105408947
M	52.754893281523415	0.1524023857803899	139.64179924118116	20.566160943094587	602.5651538894401	1241.2135331345692
M	64.4645315128236	0.1871841859760747	183.28769752768017	27.0012964422654	588.5713453708427	2244.5466387041224
M	62.67349842644208	0.1898205532436423	172.71938957413275	25.65665440958519	448.7474326496769	2004.1158388081567
M	66.17938210042385	0.1590196889819614	188.27169283040823	28.180976842786052	519.9416486377995	2499.41397975716
M	63.24650870045518	0.2007230698233208	165.96299217391706	24.32958147015172	494.1986840119411	1768.8556574874317

Final Analytical Summary

The dataset contains **5,500** valid records with a malignant diagnosis rate of **36.9%**.

The strongest feature association with malignancy is **malignancy_risk_score** with correlation **0.7764**.

The best classification model is **Random Forest** with ROC AUC **0.9999**.

The clustering section separates tumor profiles into **3** groups and reports malignancy concentration by cluster.

The anomaly-detection section flags **80** records for closer review.

The report generated **18** charts and saved all outputs in
c:\Users\nwpl\Downloads\Breast_Cancer_Data_Analysis\Output.